

Optical Transmission and Multiplexing

Outlines

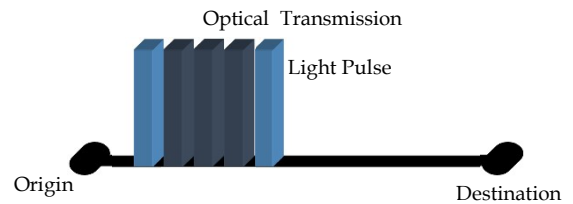
- Introduction to Optical Transmission
- Introduction to Fiber Optics Cable
- Multiplexing
- Example for Optical Transmission Networks

Introduction to Optical Transmission

- Optical Transmission
- Why Optical Transmission?
- Optical Transmission Principles

Optical Transmission

- Transmit signal/data as light-ray through an optical medium.
- Light Pulses of different wavelengths (frequencies) are used for representing 0s and 1s.



Why Optical Transmission?

- Light traveling through in a fiber is immune to electrical interference.
- It does not radiate anything.
- The fiber is less bulky than other cables.
- The signal travels longer distances.

With advanced technology:

- The cost has become very low.
- A huge amount of information can be transmitted along the single fiber strand.

Optical Transmission Principles

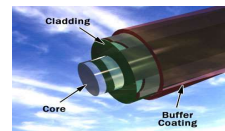
- Channel Spacing
 - The minimum frequency separation between two different signals multiplexed.
- Signal Direction
 - An optical fiber transmits signals in both uni-direction and bi direction.
- Signal Trace
 - The procedure of detecting if a signal reaches the correct destination at the other end.

Introduction to Fiber Optics Cable

- Fiber-Optics
- Fiber optics Connections
- How Fiber-optics works?
- Communication Link Between Two Computers
- Types of Fiber-optic cables
- Advantages of Fiber optics

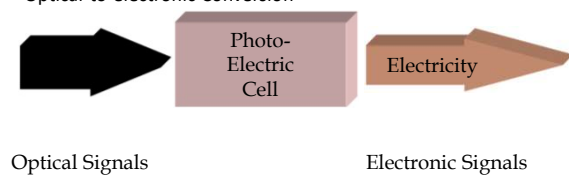
Fiber-Optics

- A fiber is a tube of glass, as thin as a hair.
- Separate its structure into two:
 - **Core** – thin glass, center of the fiber, maintains the flow of the light.
 - **Cladding** – outer optical material surrounding the core, preventing the light from escaping.
 - **Buffer Coating** – plastic coating that protects the fiber.



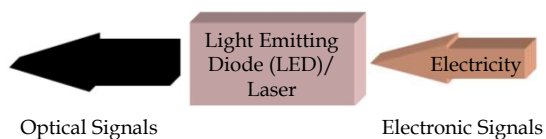
Fiber Optics Connections

- Optical-to-electronic Conversion



Fiber Optics Connections

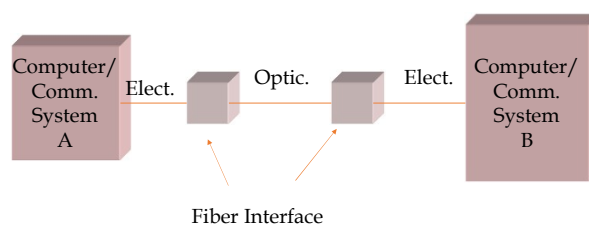
- Electronic-to-optical Conversion



How Fiber-optics works?

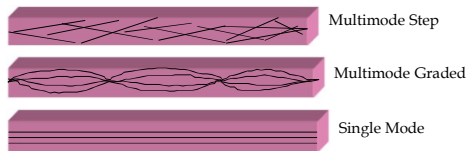
- The data to be sent is in the form of an electrical signals or a series of voltage electrical voltages represented by '0s' & '1s'.
- The '1s' & '0s' are translated into pulses of light by Light Source.
- Light pulses travels through the optical fiber.
- The Light Detector reconstitute the electrical signal form the light pulses.

Communication Link Between Two Computers



Types of Fiber optics cables

- Fibers can be divide in three major types, which are:
 - Multimode step index (for LANs):- Lower speed optical transmission
 - Multimode graded index (even better for LANs):- Intermediate speeds of transmission
 - Single mode (best for WANs):- Higher speed in transmission



Advantages of Fiber optics

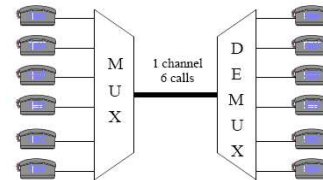
- Thinner
 - Optical fibers can be drawn to smaller diameters than copper wire.
- Higher carrying capacity
 - More fibers can be bundled into a given-diameter cable.
- Less signal degradation
 - The loss of signal in optical fiber is less than in copper wire.
- Low power
 - Because signals in optical fibers degrade less, lower-power transmitters can be used instead of the high-voltage electrical transmitters. Again, this saves your provider and you money.

Advantages of Fiber optics

- Digital signals
 - Optical fibers are ideally suited for carrying digital information, which is especially useful in computer networks.
- Non-flammable
 - Because no electricity is passed through optical fibers, there is no fire hazard.
- Lightweight
 - An optical cable weighs less than a comparable copper wire cable. Fiber-optic cables take up less space in the ground.
- Flexible
 - Because fiber optics are so flexible and can transmit and receive light, they are used in many flexible digital cameras

Multiplexing

- Multiplexing is the process of aggregating multiple low-speed channels into a single high-speed channel.
- Typically, multiplexing resides in the physical layer.
- However, multiplexing is also used in the data link layer and the transport layer.



Multiplexing Schemes

- Multiplexing schemes can be divided into four categories:
 - Frequency Division Multiplexing (FDM)
 - Wavelength Division Multiplexing (WDM)
 - Time Division Multiplexing (TDM)
 - Code Division Multiplexing (CDM)
- TDM and FDM are widely used.
- WDM is a form of FDM used for optical fiber.
- CDM is a mathematical approach used in cell phone mechanisms.

Frequency Division Multiplexing (FDM)

- Is an Analog technique.
- Transmission bandwidth is divided in frequency.
- Uses analog modulation and filtering to multiplex narrow band signals to broadband channels.
- Examples:
 - Broadcast radio and television
 - Cable television
 - The AMPS cellular phone systems

Wavelength Division Multiplexing (WDM)

- WDM refers to the application of FDM to optical fiber
- The inputs and outputs of such multiplexing are wavelengths of light.
- One most used technique is Dense WDM

Time Division Multiplexing (TDM)

- Is a digital technique.
- Transmission Bandwidth is shared on a time slot.
- This can be applied when the data rate capacity of a transmission medium is much greater than the data rate required by individual sending and receiving devices.
- Time division multiplexing comes in two basic forms:
 1. Synchronous Time Division Multiplexing.
 2. Asynchronous Time Division Multiplexing.
- Example:
 - ISDN Interface

Synchronous Time Division Multiplexing

- Data rate of medium exceeds data rate of digital signal to be transmitted
- Multiple digital signals interleaved in time
- May be at bit level or blocks
- Time slots pre-assigned to sources and fixed
- Time slots allocated even if no data
- Time slots do not have to be evenly distributed amongst sources

Asynchronous Time Division Multiplexing

- In Synchronous TDM many slots are wasted
- Statistical TDM allocates time slots dynamically based on demand
- Multiplexer scans input lines and collects data until frame full
- Data rate on line lower than aggregate rates of input lines

Code Division Multiplexing (CDM)

- Spread-spectrum communication technique.
- Transmitted signal has much wider bandwidth than information signal.
- Applications in mobile radio systems, wireless LANs, and high-speed optical fiber communication systems